

Credit, Monetary Policy, and Default Heterogeneity in Brazilian Micro and Small Enterprises (2012–2026)

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Abstract

Micro and Small Enterprises (MSEs) occupy a strategic position in the Brazilian productive structure, accounting for 50% to 55% of formal wage employment. This paper investigates how credit cycle variables and monetary conditions contribute to predicting default among MSEs across the four major economic sectors (Commerce, Construction, Manufacturing, and Services) between July 2012 and June 2026, using microdata from the Credit Information System (SCR/BCB). To overcome the arbitrariness of static thresholds, we propose an endogenous identification strategy based on discrete-time Markov Chain persistence ($P_{11} \geq 70\%$). Utilizing supervised machine learning under strict temporal validation (*Out-of-Time*, 2022–2026) and interpretability via *Tree SHAP*, results indicate that Random Forest achieved the highest overall discriminative performance (ROC-AUC of 0.8041 and PR-AUC of 0.3076 for a base prevalence of 12.96%), identifying 25 of the 28 observed critical default events. The empirical analysis reveals two primary regularities: the predictive relevance of monetary policy lags (the Selic rate with a 6-month lag) and sector-specific heterogeneity in credit dynamics. In Commerce and Manufacturing, portfolio growth displays a negative predictive association with default probability, whereas in Construction and Services the association is positive—a pattern consistent with the hypothesis of working capital deficit financing and debt rollover. Territorial analysis also highlights regional heterogeneity across Brazilian states.

Keywords: Loan Default; Micro and Small Enterprises; SCR/BACEN; Markov Chains; Random Forest; Tree SHAP; Monetary Policy.

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1 Introduction

According to official records from the Ministry of Labor and Employment (MTE / Novo CAGED / RAIS) consolidated by SEBRAE, Micro and Small Enterprises (MSEs) account for approximately 50% to 55% of formal wage employment (CLT) in Brazil. Due to this substantial contribution to job and income generation, the solvency and financial liquidity of these firms are directly linked to labor income dynamics, consumer confidence, and household financial health.

The macroeconomic relevance of MSEs also stems from their direct link to the labor market and, consequently, to household debt-servicing capacity. In an environment characterized by increasing household delinquency across the financial system, shifts in the financial health of smaller firms can generate a feedback loop connecting corporate credit, employment, and local demand. When smaller firms face liquidity distress and default on obligations, operational adjustments involving cost compression and layoffs reduce aggregate disposable income, thereby raising default probabilities on household credit.

To construct a representative proxy of national productive activity, this study delimits its empirical scope to the four major macroeconomic sectors tracked by business surveys and the National Financial System: Commerce, Construction, Manufacturing, and Services¹.

The central objective of this research is to conduct a data-driven economic investigation. By combining discrete-time Markov processes for regime identification, supervised machine learning, and non-linear interpretability methods (*Tree SHAP*), this study evaluates out-of-sample predictive performance (*Out-of-Time* — OOT) and investigates how monetary lags and structural credit asymmetries operate across MSE sectors.

The paper is organized into five sections: this Introduction, the Data Overview, the Predictive Modeling Framework, Empirical Results and Discussion, and Concluding Remarks.

2 Data Overview

2.1 Data Source, Preprocessing, and Sample

Credit microdata are sourced from the Credit Information System (SCR), managed and publicly disseminated by the Central Bank of Brazil (BCB). The raw database encompasses all lending transactions in the national financial system. A strict filter was applied to isolate Micro and Small Enterprise (MSE) credit operations, aggregated monthly by Brazilian State (Federative Unit — UF) and economic activity (CNAE), subsequently consolidated into the four major macro-sectors.

The time series cover the period from July 2012 to June 2026 (168 raw monthly observations). To construct year-over-year (YoY) percentage changes and monetary lags of up to six months ($t - 3$ and $t - 6$), the initial twelve months were reserved for covariate structuring. The final modeling sample comprises 156 effective monthly periods per sector, yielding a balanced panel of 624 monthly observations (156×4 sectors). Macro-financial control variables were extracted from the Time Series Management System (SGS/BCB): Average Monthly Selic Rate (SGS 4189), Average Working Capital Interest Rate for Non-Financial Corporations (SGS 20725), and the corresponding Banking Spread.

¹In the Services sector, due to the absence of a single aggregate category in the SCR/BACEN public microdata, we consolidated the segments of Food & Accommodation, Education, Health & Social Services, Logistics & Transportation, Administrative Services, Technical B2B Services, and Information Technology (IT).

2.2 Algebraic Definitions of Variables

Variables were constructed according to the following mathematical formulations:

$$\text{Default Rate}_{i,t} = \left(\frac{\text{Balance Overdue} > 90 \text{ Days}_{i,t}}{\text{Total Active Portfolio}_{i,t}} \right) \times 100 \quad (1)$$

$$\text{Average Loan Size}_{i,t} = \frac{\text{Total Active Portfolio}_{i,t}}{\text{Number of Active Operations}_{i,t}} \times \frac{1}{1,000} \quad (2)$$

$$\text{YoY Portfolio Growth}_{i,t} = \left(\frac{\text{Total Active Portfolio}_{i,t}}{\text{Total Active Portfolio}_{i,t-12}} - 1 \right) \times 100 \quad (3)$$

$$\text{Banking Spread}_t = \text{Working Capital Interest Rate}_t - \text{Average Selic Rate}_t \quad (4)$$

2.3 Consolidated Descriptive Statistics

Table 1 presents the consolidated descriptive statistics of the modeling dataset across the 624 monthly observations. Detailed sector-specific distributions are presented in Appendix A (Tables A1 to A4).

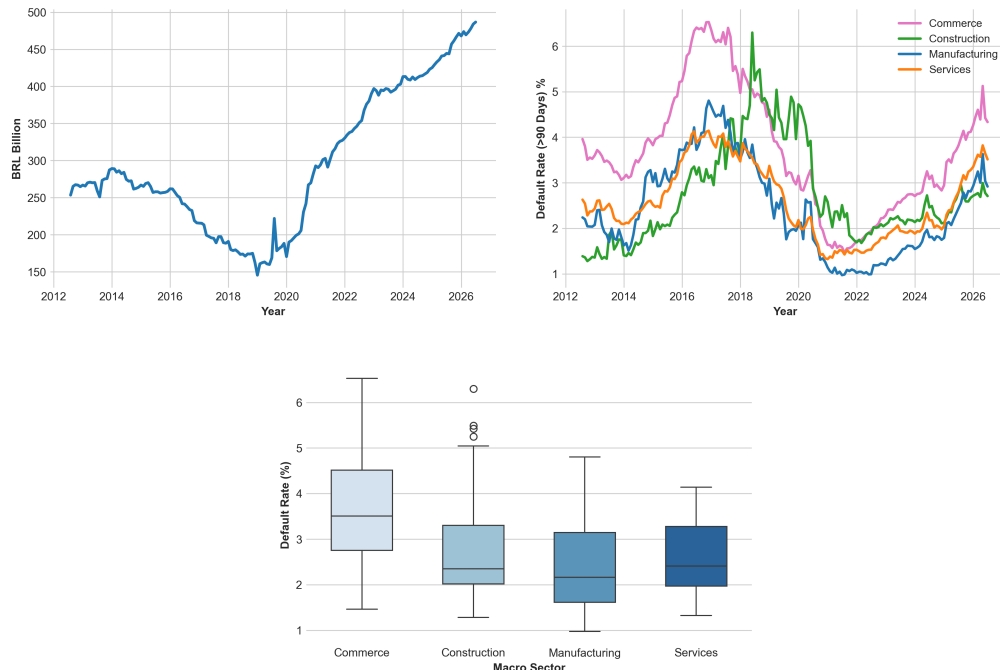
Table 1: Consolidated Descriptive Statistics (2013–2026)

Variable	Mean	Std. Dev.	Min	25% (Q1)	Median	75% (Q3)	Max
Active Portfolio (BRL Bn)	74.16	47.11	13.80	34.16	58.72	100.18	190.49
Number of Operations (Mn)	4.83	4.32	0.31	1.40	3.06	7.72	17.73
Average Ticket (BRL Th)	24.80	15.63	5.65	15.15	21.00	29.48	82.33
YoY Portfolio Growth (%)	5.42	18.90	-37.60	-7.46	4.38	15.50	83.47
Default Rate >90d (%)	2.89	1.21	0.97	1.96	2.69	3.70	6.53
Average Selic Rate (% p.a.)	10.05	3.95	1.90	6.40	10.90	13.65	14.90
Working Capital Rate (% p.a.)	20.87	4.15	10.76	17.84	21.86	23.58	29.86
Banking Spread (p.p.)	10.82	1.38	8.32	9.91	10.78	11.41	15.71

Source: Author’s elaboration based on SCR/BCB and SGS/BCB microdata. Sample size: 624 consolidated monthly observations.

The descriptive analysis reveals marked differences across sectors. Commerce (Table 3) exhibits the highest average default rate (3.69%) and the series maximum (6.53%), operating with the lowest average ticket (BRL 11.75 thousand) and the largest volume of contracts (average of 10.95 million operations). Construction (Table 4) displays the largest average loan size (BRL 47.58 thousand, reaching a maximum of BRL 82.33 thousand) and experiences default spikes concentrated in specific distress periods, peaking at 6.30% (median of 2.48%). This pattern reflects cash flow advance structures and longer contract execution horizons.

Figure 1: Historical Profile of the SCR/BCB Database (4 Sectors, 2012–2026)



Manufacturing (Table 5) registers the lowest aggregate default rate (2.44%) and rapid recovery trajectories following adverse shocks, with an average ticket of BRL 20.79 thousand. Services (Table 6) shows the highest average annual portfolio expansion (9.08% YoY), an average default rate of 2.59%, and the lowest sample dispersion (standard deviation of 0.84%). This sectoral heterogeneity demonstrates that credit expansion and default risk do not maintain a uniform relationship across industries.

2.4 Historical Trajectory and Credit Cycles

The temporal evolution of the aggregate active portfolio and default rate reveals three distinct phases over the past fourteen years.

Between 2012 and 2015, total credit remained stable between BRL 255 and BRL 260 billion (+2.0% cumulative growth relative to 2012), supported by public bank corporate lending programs distinct from household consumption credit.

Between 2016 and 2019, the domestic recession and banking sector risk aversion prompted severe balance-sheet contraction. Outstanding credit contracted by 36.5% between January 2016 and December 2019, bottoming out at BRL 145 billion (a cumulative drop of 44.2% relative to the 2014 peak), a period during which Commerce and Construction default rates surpassed 6.0%.

From 2020 onward, pandemic emergency credit guarantee facilities—specifically the National Program for Micro and Small Enterprise Support (Pronampe) and the Emergency Credit Access Program (PEAC-FGI)—alongside digital credit expansion, sparked sustained growth. The portfolio expanded by 236.6% (rising from BRL 145 billion in early 2020 to BRL 488 billion in June 2026), guided by institutional frameworks established by the Ministry of Entrepreneurship² and the National MSE Policy³.

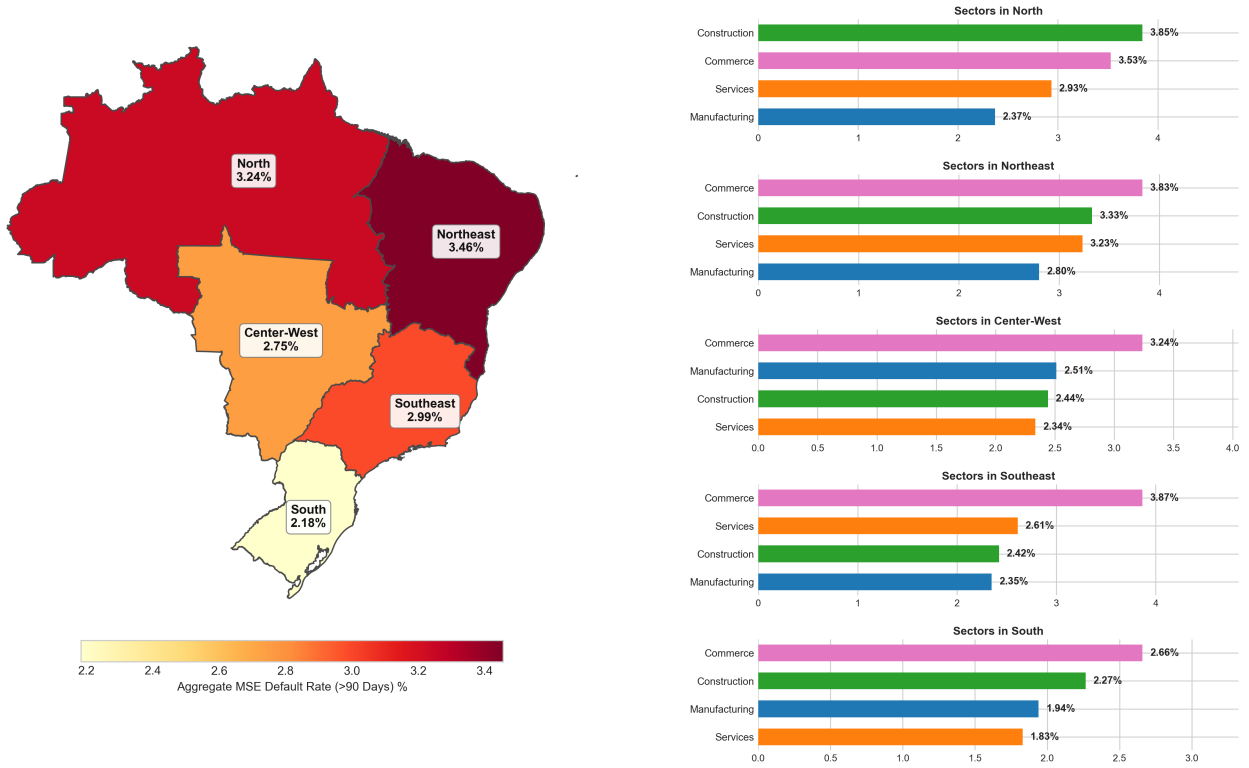
²Established by Federal Law No. 14,816/2024.

³Presidential Decree No. 11,993/2024.

2.5 Geography of Risk: Territorial and Sectoral Panel

Spatial default distributions show substantial variation across Brazilian regions. On aggregate, the Northeast (3.46%) and North (3.24%) recorded the highest historical average default rates, followed by the Southeast (2.99%), Center-West (2.75%), and South (2.18%).

Figure 2: Regional Distribution and Sectoral Asymmetry of MSE Default (SCR/BACEN)



Disaggregating by sector indicates that the highest-risk activity varies geographically. In the Northern region, Construction exhibits the highest average default rate (3.85%), ahead of Commerce (3.53%), Services (2.93%), and Manufacturing (2.37%). Conversely, in the Northeast (3.83%), Southeast (3.87%), Center-West (3.24%), and South (2.66%), Commerce consistently registers the highest average default rate.

At the state level (Table B1 in Appendix B), Commerce is the highest-risk sector across all southern and southeastern states, with Rio de Janeiro reaching 4.85% in retail credit (statewide mean of 4.10%). In the North, Construction in Acre reached the highest sectoral rate nationwide (9.94%). In the Northeast, Alagoas recorded the highest default rate in Manufacturing across all states (5.92% compared to a 4.15% state average).

3 Predictive Modeling Framework

3.1 Endogenous Risk Thresholds via Markov Regime Persistence

Defining a supervised target requires methodological rigor to prevent naive static splits (such as the historical sample median) from imposing artificial 50% class balances in training or conflating high-frequency liquidity fluctuations with structural credit distress. To establish objective cutoffs grounded in observable time-series dynamics, this study implements an endogenous threshold discovery strategy based on discrete-time Markov Chain persistence (Hamilton, 1989).

Let $y_{i,t}$ denote the MSE default rate for sector i . Over the training sample (January 2013 to December 2021), we define a binary stochastic process $\{S_{i,t}\}_{t \geq 1}$ conditional on a candidate cutoff τ :

$$S_{i,t}(\tau) = \begin{cases} 0, & \text{if } y_{i,t} < \tau \quad (\text{Normal Regime}) \\ 1, & \text{if } y_{i,t} \geq \tau \quad (\text{Structural Risk Regime}) \end{cases} \quad (5)$$

The stochastic transition probability matrix $\mathbf{P}_i(\tau)$ is expressed as:

$$\mathbf{P}_i(\tau) = \begin{pmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{pmatrix} \quad (6)$$

where $P_{jk} = \mathbb{P}(S_{i,t+1} = k \mid S_{i,t} = j)$. The expected duration in the risk state is given analytically by:

$$\mathbb{E}[D_1] = \frac{1}{1 - P_{11}} \quad (7)$$

The optimal sectoral threshold τ_i^* is obtained by solving the constrained optimization problem:

$$\tau_i^* = \arg \max_{\tau \in \Omega_i} \{P_{11}(\tau)\} \quad \text{subject to} \quad \begin{cases} P_{11}(\tau) \geq 0.70 \\ P_{00}(\tau) \geq 0.70 \end{cases} \quad (8)$$

The condition $P_{11} \geq 0.70$ ensures that the expected duration in the distress state exceeds 3.3 months ($\mathbb{E}[D_1] \geq 3.33$). This constraint guarantees that the model captures inertial and persistent credit distress regimes, filtering out intra-quarterly seasonal noise and aligning with the Central Bank’s 90-day overdue regulatory definition.

Applying this optimization over the training partition yielded the following sector-specific thresholds: Commerce ($\tau^* = 3.98\%$, $P_{11} = 97.9\%$), Construction ($\tau^* = 2.95\%$, $P_{11} = 98.0\%$), Manufacturing ($\tau^* = 3.32\%$, $P_{11} = 97.1\%$), and Services ($\tau^* = 2.65\%$, $P_{11} = 98.0\%$). Over the Out-of-Time test window (2022–2026), the observed prevalence of persistent risk months was 12.96% (28 positive events in 216 independent observations).

3.2 Out-of-Time (OOT) Experimental Setup and Metrics

To evaluate generalization and prevent temporal data leakage, a strict temporal split was implemented: training from 2013 to 2021 (102 effective months, 408 observations: 102×4 sectors) and out-of-time testing from 2022 to June 2026 (54 months, 216 independent observations).

Predictive performance was evaluated using four metrics. Accuracy measures the overall proportion of correct classifications $\frac{TP + TN}{TP + TN + FP + FN}$. The Area Under the Receiver Operating

Characteristic Curve (ROC-AUC) assesses global discriminative capability across decision thresholds (Green and Swets, 1966). The Area Under the Precision-Recall Curve (PR-AUC / *Average Precision*) evaluates minority class detection under class imbalance (Davis and Goadrich, 2006). The Brier Score assesses probability calibration relative to actual outcomes y_j (Brier, 1950):

$$\text{BS} = \frac{1}{N} \sum_{j=1}^N (\hat{p}_j - y_j)^2 \quad (9)$$

Marginal feature contributions were estimated using *Tree SHAP* (Lundberg and Lee, 2017; Shapley, 1953). Colors represent the association between feature values and their respective SHAP contributions, serving as an indicator of the predominant direction of predictive association.

Importantly, the framework classifies default conditions at period t utilizing contemporary and lagged covariates ($t - 3$ and $t - 6$), capturing predictive associations rather than structural causal effects.

4 Empirical Results and Discussion

4.1 Comparative Model Evaluation (OOT 2022–2026)

Under imbalanced class distributions (OOT prevalence of 12.96%), overall accuracy becomes an uninformative metric: a naive classifier assigning all instances to the majority class achieves 87.04% accuracy despite zero sensitivity. Consequently, regime discrimination and critical event detection are evaluated primarily through ROC-AUC and PR-AUC. Table 2 reports comparative performance across the four algorithms on the Out-of-Time test partition (2022–2026).

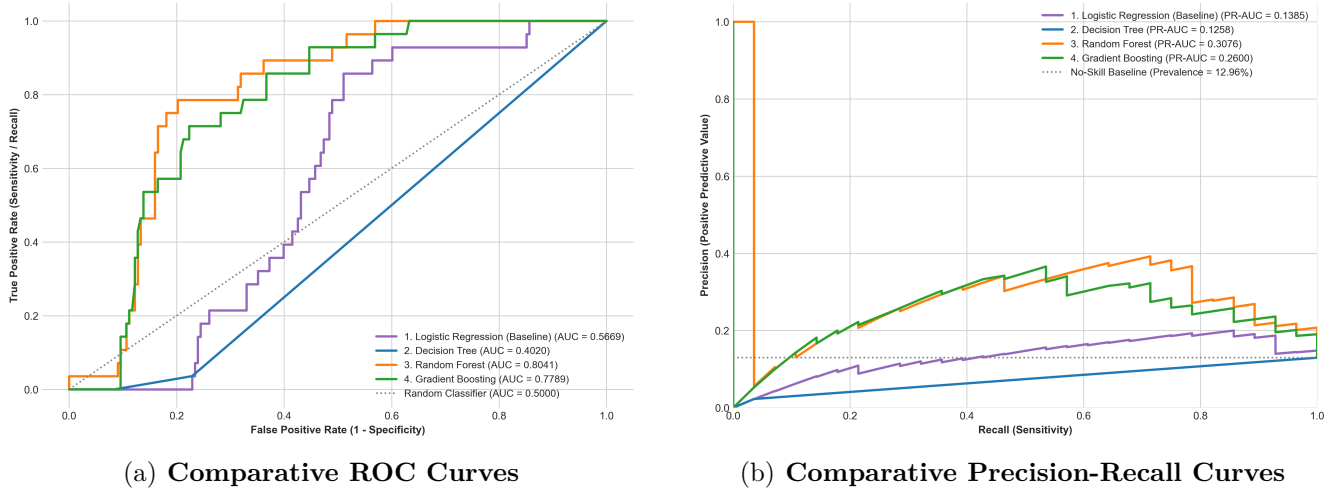
Table 2: Predictive Model Performance — Out-of-Time Validation (2022–2026)

Model	Accuracy	ROC-AUC	PR-AUC	Brier Score
1. Logistic Regression (Baseline)	82.41%	0.5669	0.1385	0.1538
2. Decision Tree	79.63%	0.4020	0.1258	0.1969
3. Random Forest	65.28%	0.8041	0.3076	0.1926
4. Gradient Boosting (GBM)	73.61%	0.7789	0.2600	0.1813

Source: Author’s elaboration. OOT test set comprising 216 monthly observations (actual prevalence of 12.96%).

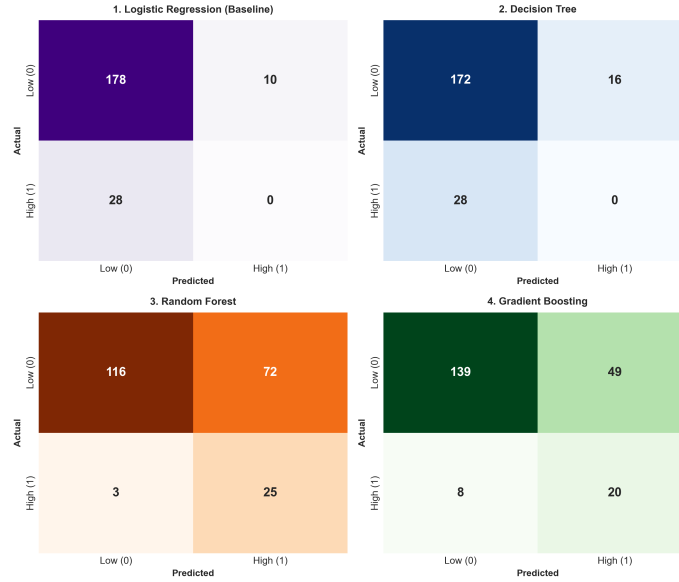
Figure 3 displays the comparative ROC and Precision-Recall Curves. Random Forest achieved the highest global discrimination, attaining a ROC-AUC of 0.8041 (compared to 0.5669 for Logistic Regression). Regarding minority class identification, the model achieved a PR-AUC of 0.3076, outperforming the 12.96% no-skill baseline and all competing classifiers.

Figure 3: Out-of-Time Discriminative Performance (OOT 2022–2026)



Additionally, Gradient Boosting produced the lowest Brier Score (0.1813), indicating competitive probability calibration. Analysis of the confusion matrices (Figure 4) shows that Random Forest successfully captured 25 of the 28 true distress events (sensitivity of 89.3%) with 72 false positives, while Gradient Boosting identified 20 events with 49 false positives, and Logistic Regression assigned all observations to the majority class.

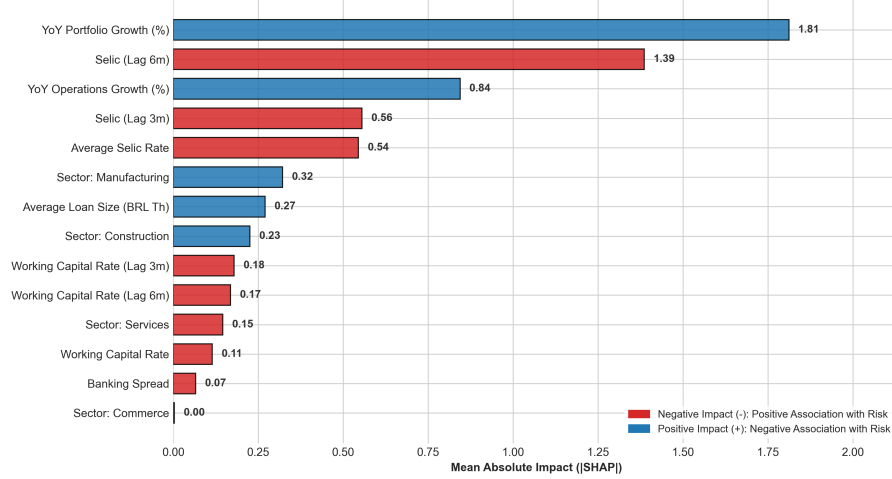
Figure 4: Comparative Confusion Matrices — Out-of-Time Test (2022–2026)



4.2 Global Feature Importance and Monetary Transmission Lags

Figure 5 summarizes the mean absolute SHAP values ($|\text{SHAP}|$) for explanatory variables in the OOT partition. The temporal dynamics confirm the predictive importance of monetary policy lags. The Selic rate lagged by six months ($\text{lag } 6$, $|\text{SHAP}| = 1.39$), Selic at three months ($\text{lag } 3$, 0.56), and contemporary Selic (0.54) all display positive associations with predicted default probability. This pattern is consistent with a monetary policy transmission lag of approximately three to six months (one to two quarters) affecting MSE financial solvency.

Figure 5: Drivers of MSE Default — Global OOT SHAP Summary



Note: Bar length reflects the mean absolute impact ($|\text{SHAP}|$), while colors indicate the predominant direction of predictive association with default probability.

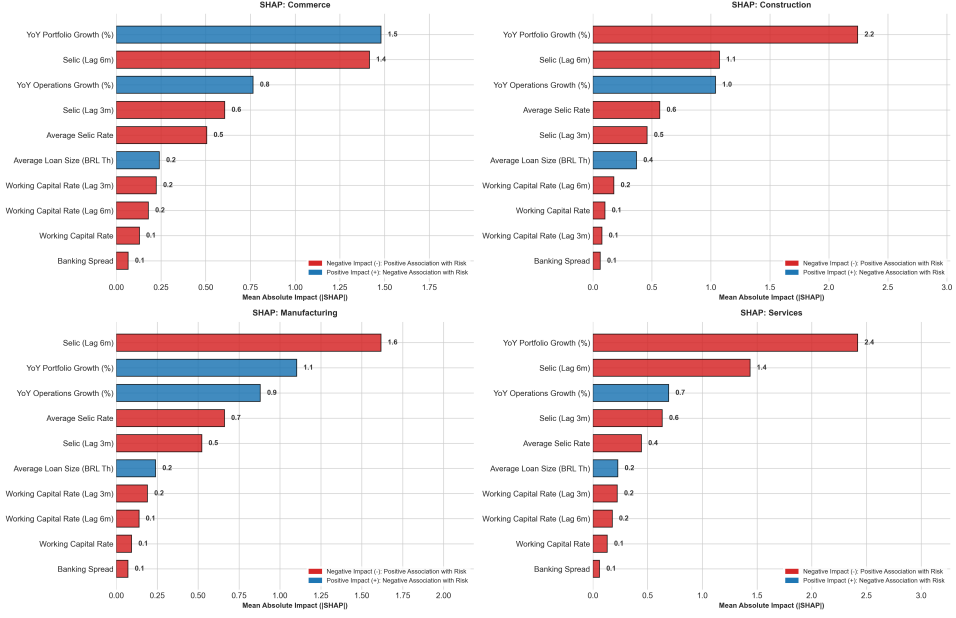
4.3 Sectoral Heterogeneity in Credit Dynamics

Disaggregating SHAP contributions across economic sectors reveals structural heterogeneity in the relationship between credit expansion and default risk.

In Commerce ($|\text{SHAP}| = 1.5$) and Manufacturing ($|\text{SHAP}| = 1.1$), higher portfolio growth is associated with lower predicted default risk. This pattern is compatible with the hypothesis that, in short-cycle and inventory-intensive sectors, credit expansion primarily supports operational working capital and liquidity.

Conversely, in Construction ($|\text{SHAP}| = 2.2$) and Services ($|\text{SHAP}| = 2.4$), the association reverses: higher credit growth is associated with increased default risk. This finding is consistent with a dynamic wherein credit growth occurs alongside the need to finance operational cash deficits or roll over existing debt. In Construction, multi-year project horizons heighten sensitivity to debt-carrying costs; in Services, limited tangible collateral amplifies financial vulnerability when new borrowing finances operating shortfalls. While this interpretation remains an economic hypothesis that cannot be directly isolated from aggregated credit variables, the relative contributions depicted in Figure 6 underscore distinct functional roles of credit across sectors.

Figure 6: Sectoral Drivers of MSE Default — Sector-Specific OOT SHAP



Note: Bar length reflects the mean absolute impact ($|\text{SHAP}|$), while colors indicate the direction of predictive association with default risk.

This sectoral divergence demonstrates that credit expansion carries distinct economic implications depending on industry cost structures and operating cycles.

4.4 Study Limitations

This study is subject to methodological limitations. First, while Markov Chain thresholding provides objective endogenous cutoffs, the target variable relies on sector-specific historical distributions rather than an exogenous balance-sheet insolvency metric. Second, out-of-time validation encompasses 216 monthly observations (with 28 positive events); thus, marginal performance differences between models should be interpreted with statistical caution. Third, SHAP contributions reflect conditional predictive associations and do not imply structural causal parameters. Finally, the analysis relies on sector- and state-level aggregations, which do not capture firm-level idiosyncratic balance-sheet heterogeneity.

5 Concluding Remarks

This paper investigated empirical determinants and predictive dynamics of default in Brazilian micro and small enterprises between 2012 and 2026, combining SCR/BCB microdata, endogenous Markov regime calibration, and supervised learning under strict temporal validation.

Three empirical regularities emerge. First, default rates exhibit marked regional and sectoral heterogeneity, with statewide averages ranging from 2.18% in the South to 3.46% in the Northeast, and Commerce concentrating the highest default rates in the South and Southeast, while Construction leads in the North. Second, the Selic rate and working capital interest rates lagged by three and six months demonstrate greater relative predictive importance than contemporary rates, consistent with a transmission lag of one to two quarters. Third, the relationship between

portfolio expansion and default risk is asymmetric: negative in Commerce and Manufacturing, but positive in Construction and Services.

These findings provide descriptive and predictive insights for risk management, macroprudential monitoring, and public credit guarantee design in emerging market enterprise lending.

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Appendices

Appendix A Descriptive Summaries by Macro-Sector (2013–2026)

Table 3: Descriptive Statistics: Commerce Sector ($N = 156$ months)

Variable	Mean	Std. Dev.	Min	25% (Q1)	Median	75% (Q3)	Max
Active Portfolio (BRL Bn)	113.32	37.54	55.89	77.86	103.78	154.87	179.48
Number of Operations (Mn)	10.95	3.32	6.56	8.28	9.20	13.81	17.73
Average Ticket (BRL Th)	11.75	6.18	5.65	6.10	7.89	18.57	21.37
YoY Portfolio Growth (%)	5.53	17.13	-26.20	-7.33	3.47	15.42	66.78
Default Rate >90d (%)	3.69	1.41	1.46	2.73	3.45	4.71	6.53

Table 4: Descriptive Statistics: Construction Sector ($N = 156$ months)

Variable	Mean	Std. Dev.	Min	25% (Q1)	Median	75% (Q3)	Max
Active Portfolio (BRL Bn)	29.29	9.84	13.80	20.96	29.75	33.79	56.06
Number of Operations (Mn)	0.64	0.24	0.31	0.46	0.55	0.82	1.23
Average Ticket (BRL Th)	47.58	12.55	30.97	37.29	45.09	58.93	82.33
YoY Portfolio Growth (%)	5.64	19.39	-31.71	-10.41	7.13	20.17	55.80
Default Rate >90d (%)	2.83	1.05	1.39	2.09	2.48	3.37	6.30

Table 5: Descriptive Statistics: Manufacturing Sector ($N = 156$ months)

Variable	Mean	Std. Dev.	Min	25% (Q1)	Median	75% (Q3)	Max
Active Portfolio (BRL Bn)	46.74	11.18	24.66	35.35	47.81	56.15	70.80
Number of Operations (Mn)	2.26	0.34	1.45	2.02	2.23	2.41	3.07
Average Ticket (BRL Th)	20.79	4.43	12.74	16.33	21.33	24.67	28.84
YoY Portfolio Growth (%)	1.42	19.14	-37.60	-9.28	0.02	8.06	67.25
Default Rate >90d (%)	2.44	1.07	0.97	1.59	2.20	3.24	4.80

Table 6: Descriptive Statistics: Services Sector ($N = 156$ months)

Variable	Mean	Std. Dev.	Min	25% (Q1)	Median	75% (Q3)	Max
Active Portfolio (BRL Bn)	107.29	43.09	48.31	71.81	85.37	151.16	190.49
Number of Operations (Mn)	5.47	1.13	3.02	4.72	5.22	5.75	9.05
Average Ticket (BRL Th)	19.07	4.87	12.46	14.64	16.83	23.14	28.98
YoY Portfolio Growth (%)	9.08	19.26	-21.00	-4.12	7.72	16.12	83.47
Default Rate >90d (%)	2.59	0.84	1.32	1.94	2.39	3.38	4.14

Appendix B Territorial Diagnostic across 27 Federative Units (UFs)

Table 7: Critical Sector of Highest Default Risk by Brazilian State (UF)

Region	State (UF)	Critical Sector	Sectoral Default (%)	Statewide Mean (%)
North	AC	Construction	9.94%	4.65%
	AM	Commerce	4.25%	3.40%
	AP	Construction	5.50%	4.38%
	PA	Construction	4.08%	3.29%
	RO	Construction	3.71%	2.66%
	RR	Construction	5.07%	3.73%
	TO	Manufacturing	3.34%	2.57%
Northeast	AL	Manufacturing	5.92%	4.15%
	BA	Commerce	3.87%	3.38%
	CE	Commerce	3.97%	3.39%
	MA	Construction	3.39%	3.23%
	PB	Construction	3.62%	3.36%
	PE	Commerce	4.32%	3.50%
	PI	Construction	4.27%	3.37%
	RN	Construction	4.48%	3.84%
	SE	Manufacturing	4.15%	3.63%
Center-West	DF	Commerce	4.59%	3.25%
	GO	Commerce	3.45%	2.96%
	MS	Construction	3.29%	2.35%
	MT	Construction	2.89%	2.37%
Southeast	ES	Commerce	2.76%	2.46%
	MG	Commerce	3.25%	2.69%
	RJ	Commerce	4.85%	4.10%
	SP	Commerce	3.95%	2.90%
South	PR	Commerce	3.12%	2.54%
	RS	Commerce	2.55%	2.16%
	SC	Commerce	2.16%	1.81%